



Gen AI and Agentic AI in Production

Week 3
Day 3

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

GitHub
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

INTRODUCING “ALEX”



Alex - the financial planner



Alex: The Agentic Learning Equities Explainer

What it is:

SaaS commercial application

Acts as a financial planner

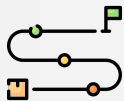
Reviews equity portfolio, makes retirement plans

Getting set up:

New repo to clone: <https://github.com/ed-donner/alex>

Time for investigation & permissions

Amazon SageMaker



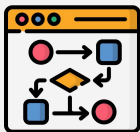
End-to-end ML development

Build, train, fine-tune, deploy, and manage your own models



Model training + hosting

Managed infrastructure for running training jobs and deploying trained models as scalable, production-ready endpoints



Designed for MLOps

Includes experiment tracking, model versioning, automated model tuning, and monitoring tools to make production ML repeatable and reliable

Bedrock vs SageMaker

Bedrock

SageMaker

Purpose

Use foundation models via API

Build, train, fine-tune, and deploy your own ML models

Models

Pre-trained models from Amazon and partners

Any model: bring your own, use open-source or prebuilt

Effort

Low - just call API, no infra to manage

Higher - manage infra, experiments, deployments

Deploy

Managed API endpoints

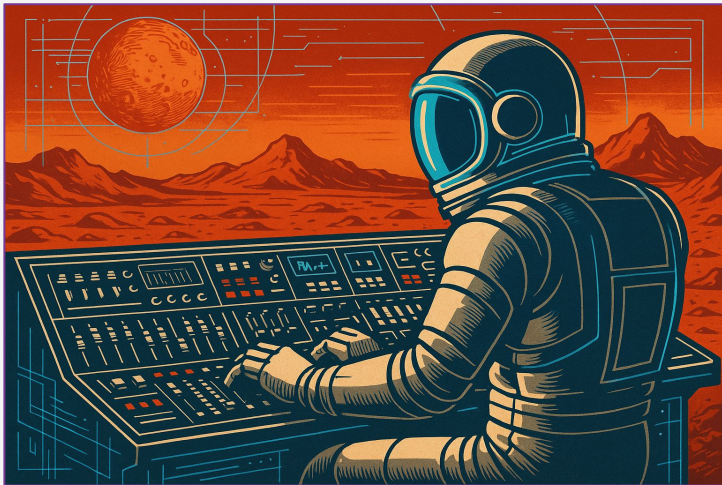
Scalable, customizable endpoints

Best for

Building with frontier models; inference at scale

End-to-end ML lifecycle, custom models

MLOps



MLOps is Devops for ML: Manage the entire ML lifecycle

Track datasets, model versions, and training configurations so you can reproduce results and roll back if needed

Monitoring & model drift – Deployed models can degrade over time as real-world data changes (“model drift”); MLOps involves monitoring predictions and retraining when accuracy drops.

SageMaker covers MLOps features: experiment tracking, model registry (for versioning), endpoints with monitoring, and automation tools for retraining when drift is detected.

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65%
Complete